

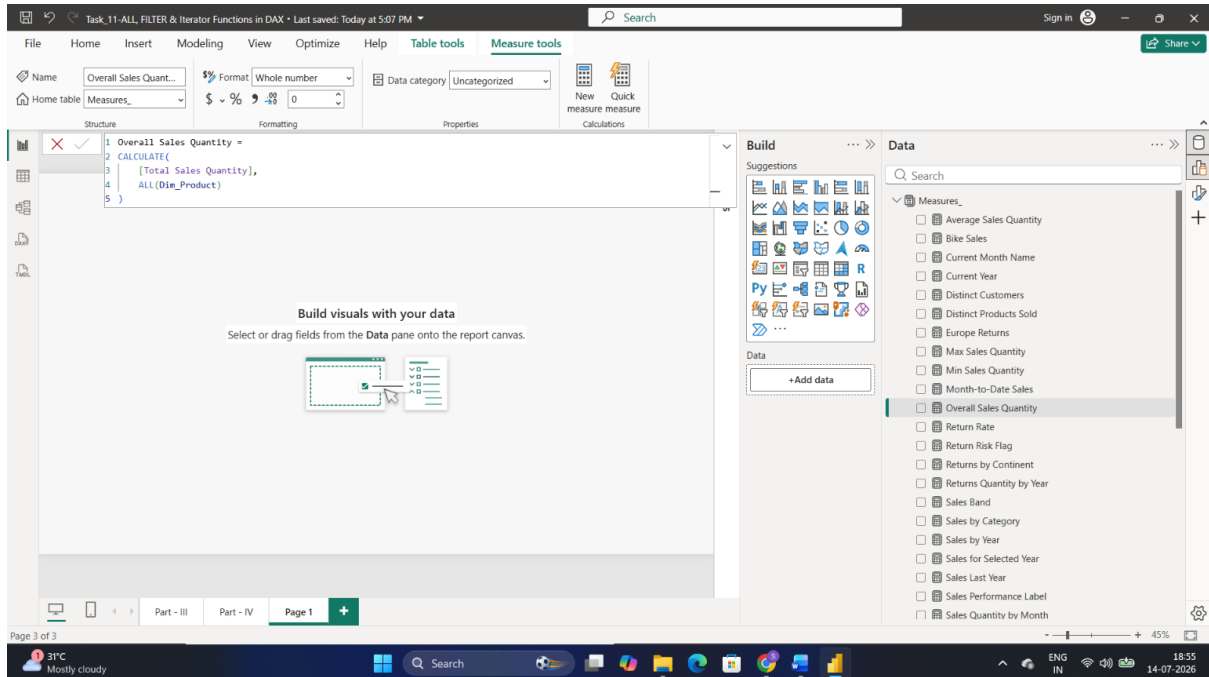
TASK: ALL, FILTER & Iterator Functions in DAX

Part 1: ALL() Function (Removing Filter Context)

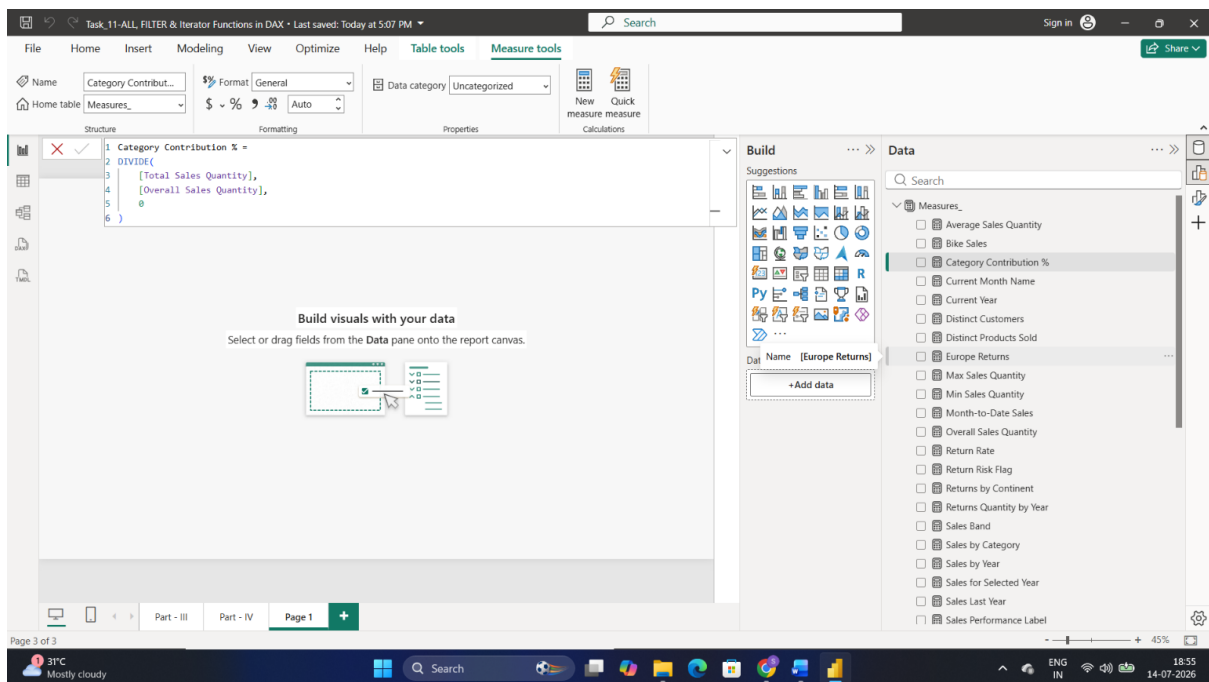
Tasks

Create the following measures:

- Overall Sales Quantity (Ignore Filters)



- Category Contribution %



TASK: ALL, FILTER & Iterator Functions in DAX

Part 2: FILTER Function (Row-Level Conditions)

Tasks

Create the following measures:

- High Volume Sales (Quantity > 5)

The screenshot shows the Microsoft Power BI Desktop interface. The 'Name' property of the measure is 'High Volume Sales'. The DAX formula is as follows:

```
1 High Volume Sales =  
2 CALCULATE(  
3     [Total Sales Quantity],  
4     FILTER(  
5         Fact_Sales,  
6         Fact_Sales[OrderQuantity] > 5  
7     )  
8 )
```

The 'Data' pane on the right shows the 'Measures' list with 'High Volume Sales' selected. The 'Build' pane on the left shows the 'Build visuals with your data' instructions.

- High Price Returns

The screenshot shows the Microsoft Power BI Desktop interface. The 'Name' property of the measure is 'High Price Returns'. The DAX formula is as follows:

```
1 High Price Returns =  
2 CALCULATE(  
3     [Total Return Quantity],  
4     FILTER(  
5         Dim_Product,  
6         Dim_Product[ProductPrice] > 1000  
7     )  
8 )
```

The 'Data' pane on the right shows the 'Measures' list with 'High Price Returns' selected. The 'Build' pane on the left shows the 'Build visuals with your data' instructions.

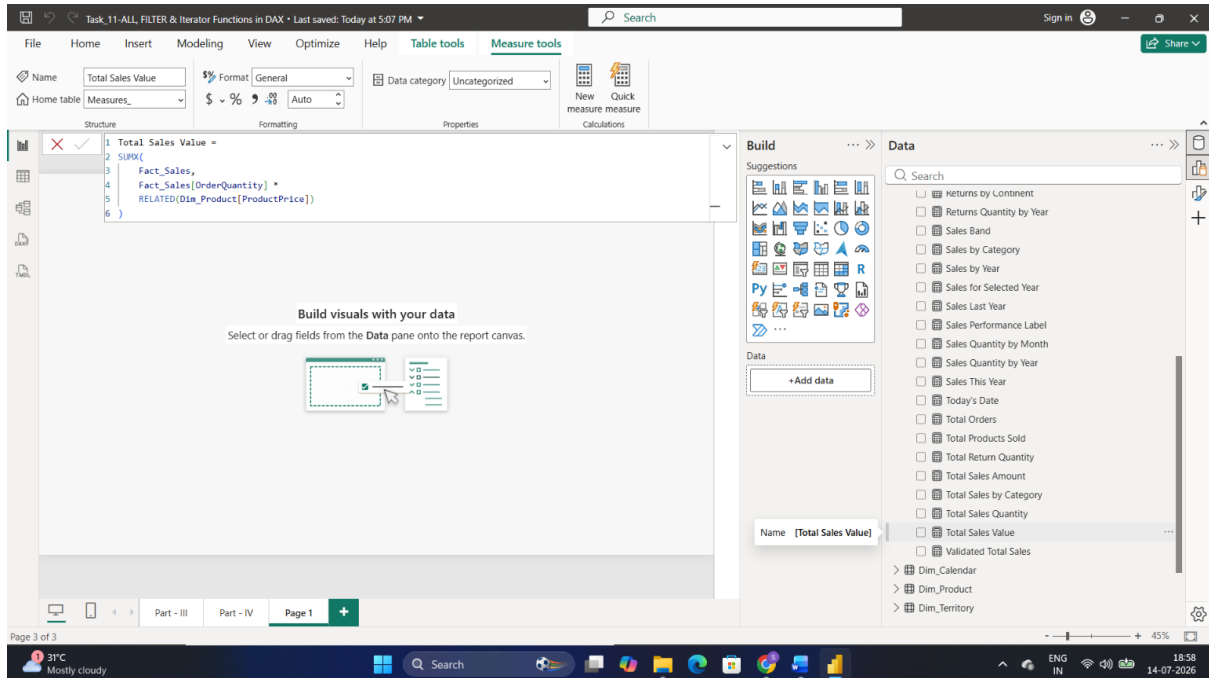
Part 3: Iterator Functions (SUMX, AVERAGEX)

TASK: ALL, FILTER & Iterator Functions in DAX

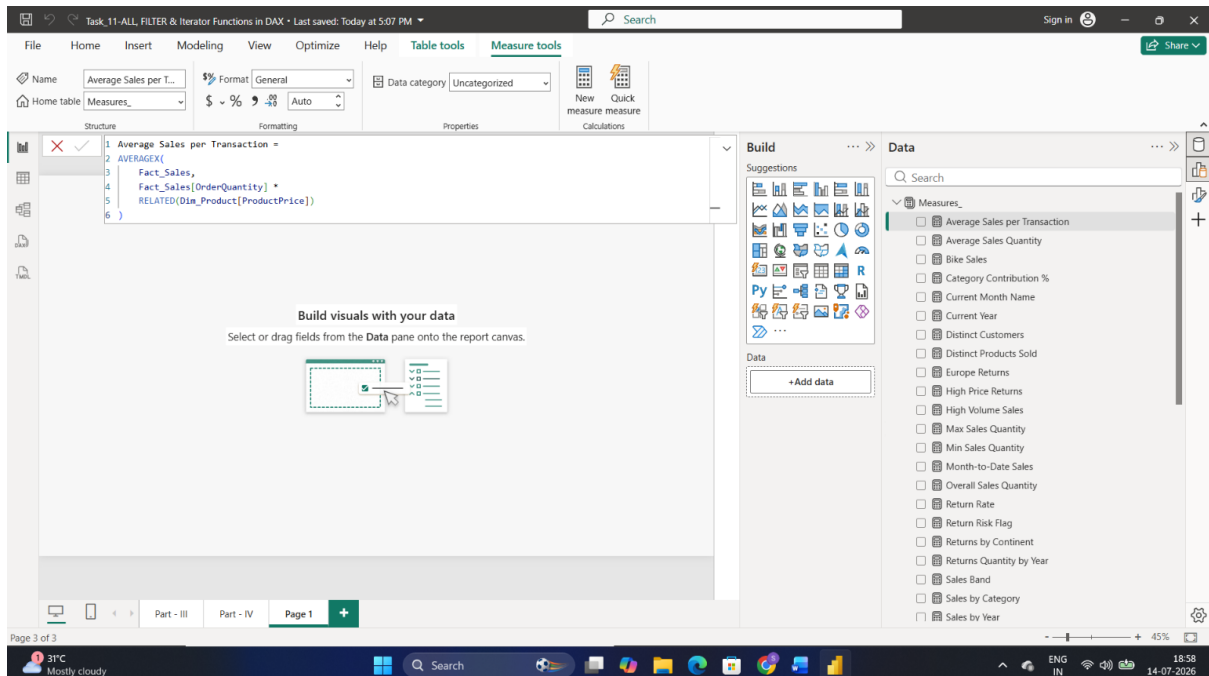
Tasks

Create the following measures:

- Total Sales Value (Quantity × Price)



- Average Sales per Transaction



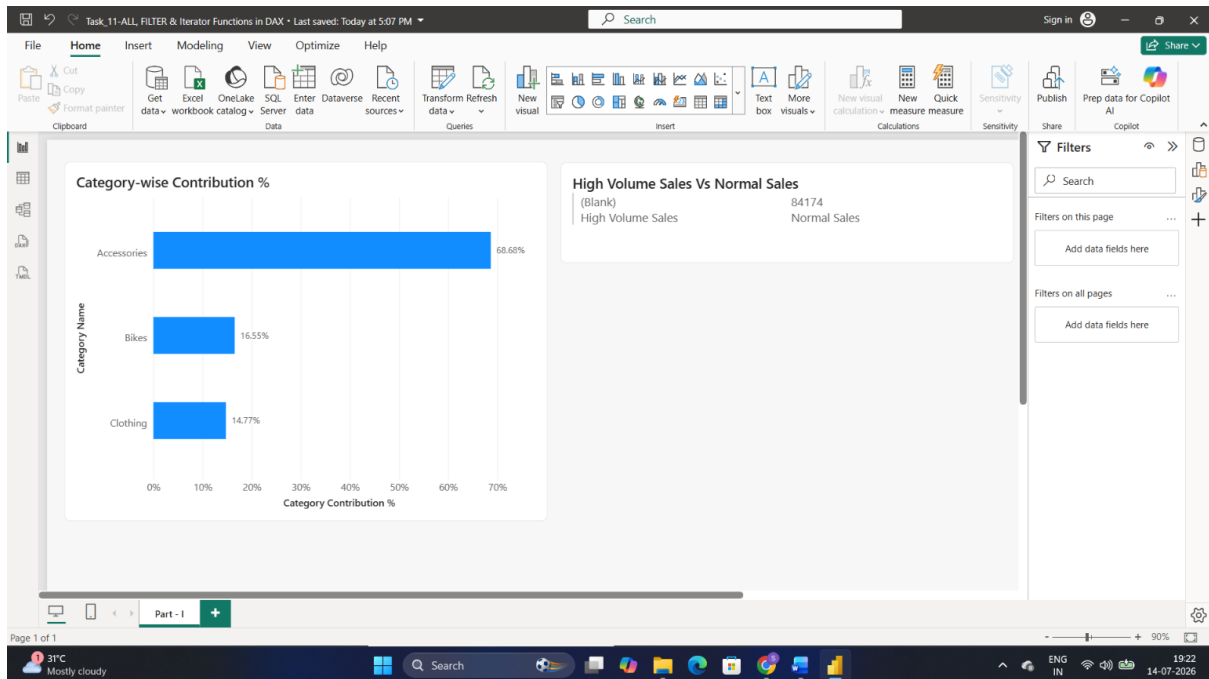
Part 4: Business Insights Using Advanced DAX

Tasks

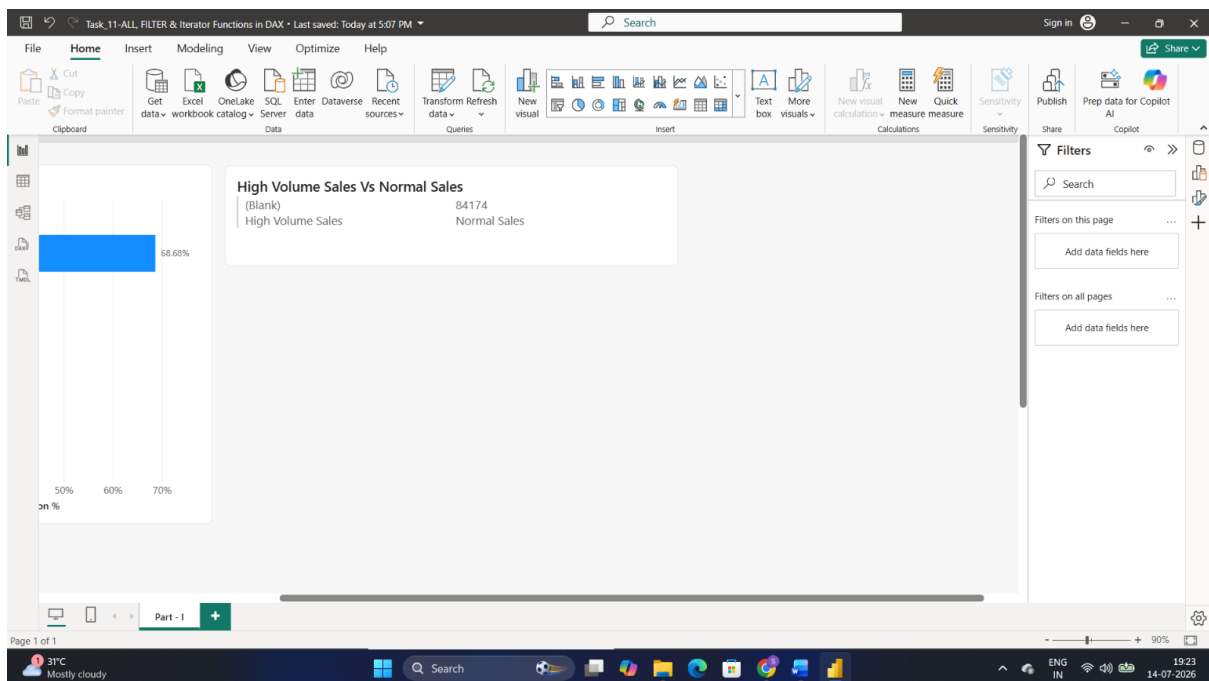
TASK: ALL, FILTER & Iterator Functions in DAX

Create visuals to answer:

- Category-wise contribution %



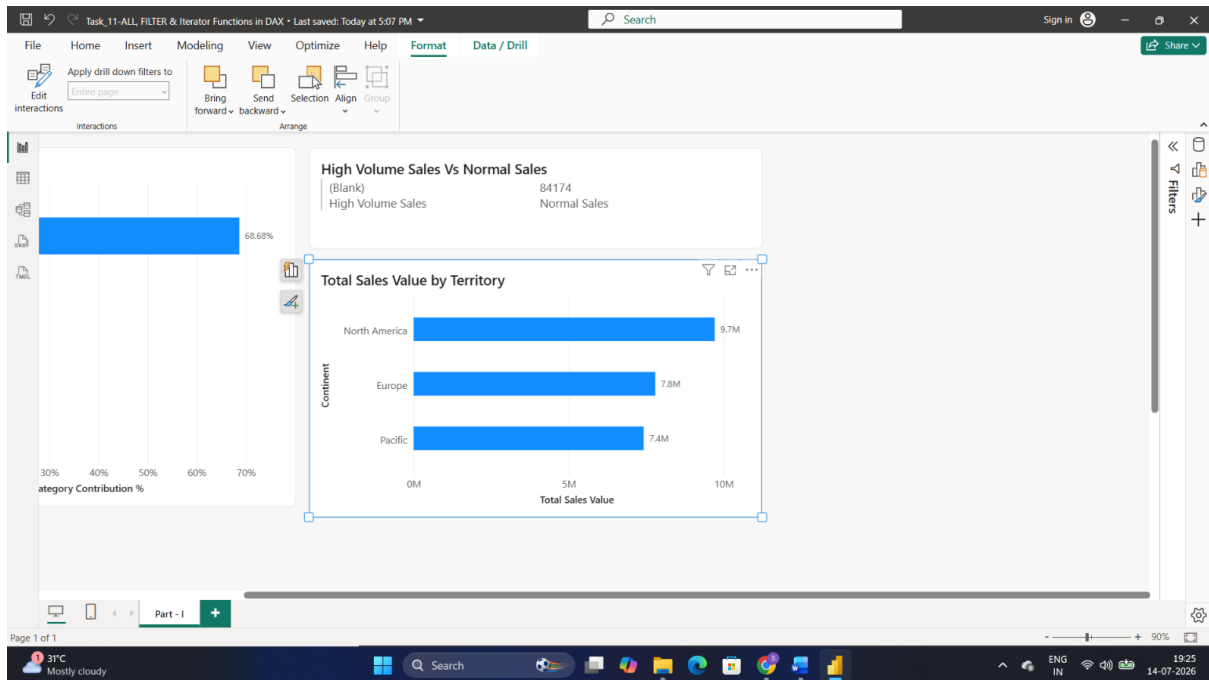
- High volume vs normal sales comparison



No sales transactions satisfy the High Volume Sales condition (Order Quantity > 5). Therefore, all recorded sales (84,174 units) fall under Normal Sales, indicating that sales are primarily driven by lower-quantity orders.

- Sales value by territory

TASK: ALL, FILTER & Iterator Functions in DAX



Analytical Questions

Q1. Which category contributes the most to overall sales?

Accessories contributes the most to overall sales, accounting for approximately **68.68%** of the total sales quantity. This makes Accessories the leading product category in terms of sales contribution.

Q2. Why is ALL mandatory for percentage calculations?

The **ALL()** function is mandatory for percentage calculations because it removes the current filter context and returns the overall total. This allows each category's sales to be compared against the grand total. Without **ALL()**, the denominator would be filtered to the current category, causing the percentage to incorrectly return **100%** for every category instead of showing the true contribution percentage.